OXALIPLATIN

Cold-induced neuropathy

Glucose 5% only

CHEMICAL NAME [(1R,2R)-cyclohexane-1,2-diamine-κ²N,N′][ethanedioato(2−)-κ²O¹,O²]platinum(II)

CLASS Third-generation platinum cytotoxic agent

TRADE NAMES Eloxatin · Oxaliplatin injection · Oxitan

THERAPEUTIC USES / INDICATIONS

- Colorectal cancer (adjuvant and metastatic: FOLFOX, CAPOX).
- Gastric, oesophageal, pancreatic (FOLFIRINOX) cancers.

ADMINISTRATION

- **IV** infusion over 2–6 h in glucose 5% only (incompatible with saline/chloride and alkaline solutions).
- Flush line with glucose 5% before other drugs (e.g. 5-FU).
- Avoid cold drinks and cold exposure during and days after infusion.

DOSAGES

- **FOLFOX:** 85 mg/m² IV every 2 weeks.
- **CAPOX:** 130 mg/m² every 3 weeks with capecitabine.
- Dose reduction/delay for neuropathy and cytopenias.

PHARMACOKINETICS

- Rapid non-enzymatic transformation to active platinum species.
- Excreted in urine. **t½:** terminal ~270 h.

NURSING MANAGEMENT

- Use glucose 5% only.
- Assess neuropathy (grading) before each cycle.
- Warn about cold sensitivity; provide gloves; avoid ice chips (unlike 5-FU bolus mucositis advice).
- Monitor for hypersensitivity; FBC before cycles.

PHARMACODYNAMICS

- Forms **platinum–DNA adducts** → apoptosis; active in cisplatin-resistant tumours.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Acute cold-induced dysaesthesia: tingling of hands, feet, throat (pharyngolaryngeal spasm — feeling unable to breathe/swallow).
- Nausea, diarrhoea, fatigue.
- Taste changes.

ADVERSE EFFECTS (SERIOUS)

- Cumulative sensory peripheral neuropathy (may be permanent).
- Myelosuppression.
- Hypersensitivity/anaphylaxis.
- Hepatic sinusoidal obstruction; interstitial lung disease; PRES.

CONTRAINDICATIONS & CAUTIONS

- Peripheral sensory neuropathy with functional impairment before first course.
- Severe renal impairment (CrCl <30).
- Myelosuppression; pregnancy.

KEY INTERACTIONS

- **5-FU:** synergistic (given together).
- **Chloride solutions:** degradation.
- **QT drugs:** caution (QT prolongation reported).

PATIENT EDUCATION

- Avoid cold drinks, cold foods and touching cold objects for 3–5 days after treatment; wear gloves and scarf in cold weather.
- Throat tightness with cold is a temporary nerve effect — stay calm, warm drinks help.
- Report numbness that affects buttoning or walking.
- Report fever ≥38 °C.

KEY POINTS

- Platinum agent for colorectal cancer (FOLFOX).
- Glucose 5% only.
- Cold-induced neuropathy; avoid cold.
- Cumulative neuropathy limits dose.

SOURCES CHECKED **[878]** Oxaliplatin 5 mg/ml concentrate for solution for infusion (UK SmPC) **[879]** Oxaliplatin, IV (US label) **[880]** Oxaliplatin, IV (US label, Qilu Pharmaceutical Co)

Full references p. 432–457